# Assignment 0 - Software Lab (DBMS)

Monthly Sales Report Generation

Mrigaj Shaw

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## Contents

# 1 Data Generation Script - `generate_data.py`

This Python script generates the synthetic sales dataset (`sales_data.csv`) with 1500 rows of randomised sales records across 4 regions, 6 salesmen, and 8 product codes.

```python
import random
import csv

random.seed(22)

regions = [1,2,3,4]
salesmen = [1,2,3,4,5,6]
products = ['A1010000','B2000000','C3000000','D4000000','E5000000','F6000000','G7000000','H8000000']

n = 1500

rows = []

for _ in range(n):
    region = random.choice(regions)
    salesman = random.choice(salesmen)
    product = random.choice(products)

    units = random.choice(
            [random.randint(1, 10)] * 6 + [random.randint(11,20)] * 3 +
            [random.randint(21,35)] * 1
    )

    rows.append((region, salesman, product, units))

random.shuffle(rows)

with open('sales_data.csv', 'w') as f:
    writer = csv.writer(f)
    writer.writerow(['Region no','Salesman no','Product Code','Units Sold'])
    writer.writerows(rows);
```

Listing 1: generate_data.py

## 2 Report Generator - `report.cpp`

This C++ program reads the CSV data, computes the total sales amount per salesman per region using predefined product prices, and writes a formatted report to `report.txt`.

```cpp
#include <ctime> // Required for generating the timestamp
#include <fstream>
#include <iomanip>
#include <iostream>
#include <string>

using namespace std;

int main() {
    float totals[4][6] = {0}; // 4 regions, each with six salesmen
    ifstream inFile("./sales_data.csv");
    ofstream outFile("report.txt");

    if (!inFile.is_open()) {
        cerr << "Error : Couldn't open data file" << endl;
        return -1;
    }

    string header;
    getline(inFile, header);

    float ProductPrices[8][2] = {
        {0, 350.00}, {1, 200.00}, {2, 178.20}, {3, 199.99},
        {4, 457.24}, {5, 599.00}, {6, 999.99}, {7, 800.00},
    };

    int r, s, u;
    string p;
    string token;

    while (getline(inFile, token, ',')) {
        r = stoi(token);

        getline(inFile, token, ',');
        s = stoi(token);

        getline(inFile, token, ',');
        p = token;

        getline(inFile, token);
        u = stoi(token);

        if (r >= 1 && r <= 4 && s >= 1 && s <= 6) {
            char productChar = p[p.find_first_not_of(" ")];
            int productIndex = productChar - 'A';

            float price = ProductPrices[productIndex][1];
            totals[r - 1][s - 1] += price * u;
        }
    }

    inFile.close();

```

```cpp
    time_t now = time(nullptr);
    tm *localTime = localtime(&now);

    outFile << "============================================\n";
    outFile << "               ABC COMPANY                  \n";
    outFile << "          MONTHLY SALES REPORT              \n";
    outFile << "============================================\n";
    outFile << "Generated on: " << put_time(localTime, "%B %d, %Y at %I:%M %p")
            << "\n\n";

    for (int i = 0; i < 4; i++) {
        outFile << "|                REGION " << (i + 1)
                << "                  |\n";
        outFile << "+------------------------------------------+\n";
        outFile << left << setw(20) << "  Salesman" << " | " << right
                << setw(18) << "Sales Amount  " << "\n";
        outFile << "--------------------------------------------\n";

        float regionTotal = 0.00;
        for (int j = 0; j < 6; j++) {
            outFile << "  Salesman " << left << setw(9) << (j + 1) << "| "
                    << "Rs. " << right << setw(10) << fixed << setprecision(2)
                    << totals[i][j] << "/-\n";
            regionTotal += totals[i][j];
        }
        outFile << "--------------------------------------------\n";
        outFile << left << setw(21) << "  TOTAL SALES" << "| "
                << "Rs. " << right << setw(10) << regionTotal << "/-\n";
        outFile << "+------------------------------------------+\n";
        outFile << right << setw(29) << "- Page " << (i + 1) << " -\n\n";
    }

    outFile.close();

    cout << "Report generated successfully!\n";

    return 0;
}
```

Listing 2: report.cpp

# 3 Generated Report Output - `report.txt`

The following is the output produced by the compiled C++ program. It shows the monthly sales figures grouped by region, with per-salesman breakdowns and region totals.

```
=============================================
                ABC COMPANY
           MONTHLY SALES REPORT
=============================================
Generated on: July 22, 2026 at 07:08 PM

|                  REGION 1                 |
+-------------------------------------------+
  Salesman           |     Sales Amount
---------------------------------------------
  Salesman 1        | Rs.  358681.19/-
  Salesman 2        | Rs.  469890.22/-
  Salesman 3        | Rs.  353507.31/-
  Salesman 4        | Rs.  332256.59/-
  Salesman 5        | Rs.  313687.59/-
  Salesman 6        | Rs.  375041.22/-
---------------------------------------------
  TOTAL SALES        | Rs. 2203064.25/-
+-------------------------------------------+
                      - Page 1 -


|                  REGION 2                 |
+-------------------------------------------+
  Salesman           |     Sales Amount
---------------------------------------------
  Salesman 1        | Rs.  289336.09/-
  Salesman 2        | Rs.  275677.44/-
  Salesman 3        | Rs.  342709.31/-
  Salesman 4        | Rs.  267423.00/-
  Salesman 5        | Rs.  264981.97/-
  Salesman 6        | Rs.  244009.70/-
---------------------------------------------
  TOTAL SALES        | Rs. 1684137.50/-
+-------------------------------------------+
                      - Page 2 -


|                  REGION 3                 |
+-------------------------------------------+
  Salesman           |     Sales Amount
---------------------------------------------
  Salesman 1        | Rs.  382560.44/-
  Salesman 2        | Rs.  246537.98/-
  Salesman 3        | Rs.  270443.78/-
  Salesman 4        | Rs.  343691.91/-
  Salesman 5        | Rs.  368745.84/-
  Salesman 6        | Rs.  341525.75/-
---------------------------------------------
  TOTAL SALES        | Rs. 1953505.75/-
+-------------------------------------------+
                      - Page 3 -

```

```

|                    REGION 4                    |
+------------------------------------------------+
  Salesman             |      Sales Amount
------------------------------------------------
  Salesman 1          | Rs.  454504.50/-
  Salesman 2          | Rs.  304662.97/-
  Salesman 3          | Rs.  366650.50/-
  Salesman 4          | Rs.  324155.38/-
  Salesman 5          | Rs.  388050.88/-
  Salesman 6          | Rs.  365744.94/-
------------------------------------------------
  TOTAL SALES          | Rs. 2203769.25/-
+------------------------------------------------+
                        - Page 4 -
```

Listing 3: report.txt - Program Output

# 4 Sales Dataset - `sales_data.csv` (Top 100 Rows)

The complete dataset contains 1500 rows. The first 100 rows are shown below.

| # | Region No | Salesman No | Product Code | Units Sold |
|---|---|---|---|---|
| 1 | 4 | 3 | D4000000 | 10 |
| 2 | 2 | 1 | E5000000 | 8 |
| 3 | 4 | 5 | G7000000 | 13 |
| 4 | 3 | 1 | F6000000 | 8 |
| 5 | 4 | 1 | H8000000 | 10 |
| 6 | 3 | 5 | G7000000 | 11 |
| 7 | 4 | 6 | B2000000 | 9 |
| 8 | 1 | 5 | B2000000 | 9 |
| 9 | 2 | 1 | G7000000 | 14 |
| 10 | 1 | 2 | H8000000 | 10 |
| 11 | 2 | 3 | A1010000 | 19 |
| 12 | 2 | 5 | A1010000 | 28 |
| 13 | 1 | 6 | B2000000 | 8 |
| 14 | 1 | 4 | G7000000 | 10 |
| 15 | 1 | 3 | F6000000 | 3 |
| 16 | 1 | 6 | B2000000 | 3 |
| 17 | 3 | 1 | A1010000 | 7 |
| 18 | 4 | 6 | G7000000 | 5 |
| 19 | 3 | 6 | G7000000 | 18 |
| 20 | 1 | 2 | G7000000 | 9 |
| 21 | 4 | 2 | G7000000 | 9 |
| 22 | 2 | 5 | H8000000 | 11 |
| 23 | 4 | 5 | G7000000 | 8 |
| 24 | 4 | 2 | G7000000 | 2 |
| 25 | 1 | 1 | B2000000 | 4 |
| 26 | 2 | 5 | C3000000 | 19 |
| 27 | 1 | 5 | D4000000 | 13 |
| 28 | 1 | 6 | F6000000 | 10 |
| 29 | 1 | 4 | H8000000 | 15 |
| 30 | 3 | 5 | B2000000 | 13 |
| 31 | 1 | 3 | G7000000 | 2 |
| 32 | 4 | 5 | A1010000 | 9 |
| 33 | 4 | 4 | G7000000 | 10 |
| 34 | 1 | 6 | G7000000 | 14 |
| 35 | 4 | 4 | H8000000 | 6 |
| 36 | 2 | 1 | A1010000 | 8 |
| 37 | 3 | 4 | A1010000 | 10 |
| 38 | 2 | 1 | E5000000 | 6 |
| 39 | 4 | 5 | D4000000 | 3 |
| 40 | 4 | 1 | G7000000 | 15 |
| 41 | 3 | 3 | B2000000 | 8 |
| 42 | 1 | 5 | E5000000 | 7 |
| 43 | 1 | 2 | G7000000 | 17 |

*Continued on next page...*

| # | Region No | Salesman No | Product Code | Units Sold |
|---|---|---|---|---|
| 44 | 1 | 2 | B2000000 | 8 |
| 45 | 2 | 3 | B2000000 | 4 |
| 46 | 3 | 2 | A1010000 | 7 |
| 47 | 2 | 2 | B2000000 | 13 |
| 48 | 3 | 3 | A1010000 | 12 |
| 49 | 3 | 5 | G7000000 | 15 |
| 50 | 4 | 2 | F6000000 | 24 |
| 51 | 4 | 5 | E5000000 | 18 |
| 52 | 4 | 3 | H8000000 | 17 |
| 53 | 3 | 1 | B2000000 | 1 |
| 54 | 4 | 2 | B2000000 | 6 |
| 55 | 2 | 6 | B2000000 | 5 |
| 56 | 3 | 2 | B2000000 | 10 |
| 57 | 2 | 5 | F6000000 | 17 |
| 58 | 1 | 2 | F6000000 | 5 |
| 59 | 2 | 2 | B2000000 | 16 |
| 60 | 1 | 2 | G7000000 | 9 |
| 61 | 3 | 5 | D4000000 | 17 |
| 62 | 4 | 5 | F6000000 | 3 |
| 63 | 4 | 4 | G7000000 | 10 |
| 64 | 1 | 3 | A1010000 | 8 |
| 65 | 4 | 5 | G7000000 | 12 |
| 66 | 1 | 5 | A1010000 | 14 |
| 67 | 2 | 3 | G7000000 | 7 |
| 68 | 1 | 2 | F6000000 | 15 |
| 69 | 1 | 6 | A1010000 | 10 |
| 70 | 4 | 1 | E5000000 | 13 |
| 71 | 3 | 1 | E5000000 | 27 |
| 72 | 1 | 3 | B2000000 | 13 |
| 73 | 4 | 2 | E5000000 | 4 |
| 74 | 2 | 1 | H8000000 | 31 |
| 75 | 1 | 3 | F6000000 | 18 |
| 76 | 4 | 3 | D4000000 | 8 |
| 77 | 3 | 5 | D4000000 | 29 |
| 78 | 1 | 5 | D4000000 | 5 |
| 79 | 1 | 5 | H8000000 | 8 |
| 80 | 3 | 4 | H8000000 | 5 |
| 81 | 3 | 5 | C3000000 | 7 |
| 82 | 4 | 6 | G7000000 | 3 |
| 83 | 1 | 5 | C3000000 | 8 |
| 84 | 4 | 3 | A1010000 | 11 |
| 85 | 3 | 6 | F6000000 | 15 |
| 86 | 2 | 6 | E5000000 | 16 |
| 87 | 4 | 6 | H8000000 | 17 |
| 88 | 1 | 2 | C3000000 | 6 |
| 89 | 1 | 3 | A1010000 | 8 |
| 90 | 4 | 4 | A1010000 | 13 |
| 91 | 4 | 2 | A1010000 | 5 |

*Continued on next page...*

| # | Region No | Salesman No | Product Code | Units Sold |
|---|---|---|---|---|
| 92 | 4 | 6 | G7000000 | 9 |
| 93 | 1 | 2 | E5000000 | 8 |
| 94 | 4 | 3 | H8000000 | 7 |
| 95 | 4 | 5 | A1010000 | 14 |
| 96 | 4 | 4 | C3000000 | 19 |
| 97 | 2 | 3 | G7000000 | 22 |
| 98 | 3 | 2 | D4000000 | 5 |
| 99 | 3 | 2 | G7000000 | 16 |
| 100 | 1 | 5 | D4000000 | 2 |

*Note: The full dataset contains 1500 rows. Only the first 100 rows are displayed here. The complete file is included as `sales_data.csv`.*